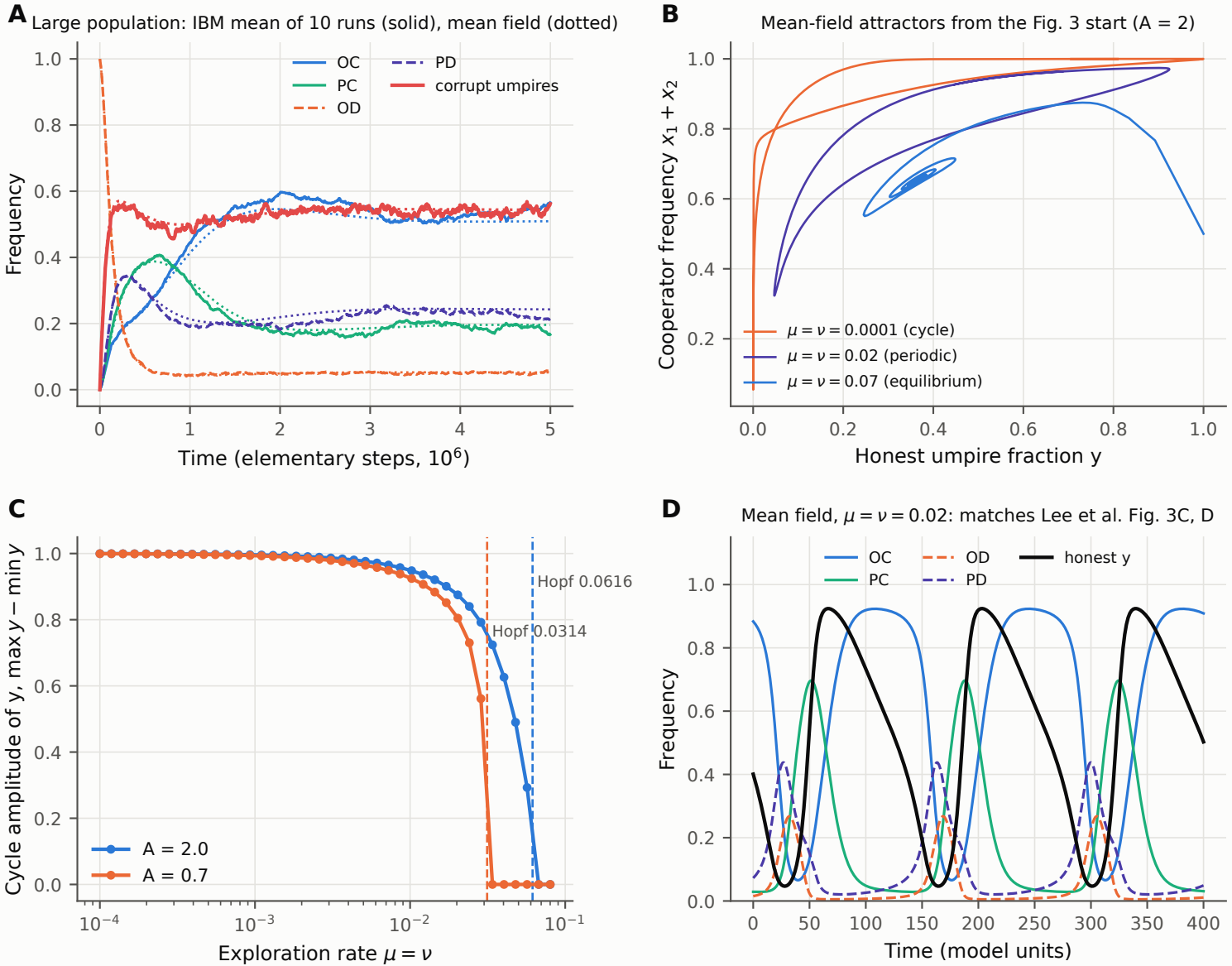

A
Large population: IBM mean of 10 runs (solid), mean field (dotted)
OC
PC
OD
PD
corrupt umpires
Frequency
Time (elementary steps, 10^6)
B
Mean-field attractors from the Fig. 3 start (A = 2)
Cooperator frequency $x_1 + x_2$
Honest umpire fraction y
$\mu = \nu = 0.0001$ (cycle)
$\mu = \nu = 0.02$ (periodic)
$\mu = \nu = 0.07$ (equilibrium)
C
Cycle amplitude of y, max y − min y
Hopf 0.0616
Hopf 0.0314
A = 2.0
A = 0.7
Exploration rate $\mu = \nu$
D
Mean field, $\mu = \nu = 0.02$: matches Lee et al. Fig. 3C, D
OC
PC
OD
PD
honest y
Frequency
Time (model units)